# Reduction #4: The Gaussian-Process / Frozen-Kernel Map and Capacity Saturation as the Synaptic Rise-Time Vanishes

Tempotron learning-surface program

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**Abstract**

We recast the tempotron's membrane-potential trace as a stationary Gaussian process (GP) over time, with covariance set by the autocorrelation of the double-exponential synaptic kernel $K(s) = V_0\big(e^{-s/\tau} - e^{-s/\tau_s}\big)\Theta(s)$. The kernel eigenspectrum—a double-Lorentzian power spectral density—yields, via the Whittle/Szegő approximation, an explicit effective dimension

$$K_{\text{eff}}^{\text{PR}}(\tau, \tau_s, T) \;=\; \frac{T(\tau+\tau_s)}{\tau^2 + 3\tau\tau_s + \tau_s^2},$$

defined as the participation ratio of the Karhunen–Loève (KL) spectrum. This spectral count replaces the abstract bin count $K$ in the capacity lift $\alpha_c = (\ln\ln K_{\text{eff}})/(2\ln 2)$. Its headline prediction—*capacity saturates as $\tau_s \to 0$ rather than diverging*—directly contradicts the up-crossing-count extrapolation $K_{\text{up}} = T/(\pi\sqrt{\tau\tau_s}) \to \infty$ of Rubin, Monasson, and Sompolinsky. We confirm the saturation by two independent routes: the participation-ratio derivation and a codex OU-limit cross-check, which shows that as $\tau_s \to 0$ the membrane GP converges to an Ornstein–Uhlenbeck process whose extremal statistics obey Berman/Pickands asymptotics $\mathbb{E}[\max V] \sim \sqrt{2\ln(T/\tau)}$—the participation-ratio scale. Status: **PARTIAL**—spectral results solid and doubly verified; the identification of the participation ratio as the argument of the replica saddle lift is a physically motivated hypothesis not yet derived from first principles.

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*Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.*

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# 1 Introduction

The tempotron [4] is a leaky integrate-and-fire neuron whose membrane potential, driven by an $N$-afferent spike-timing pattern, is

$$V(t;\mathbf{w}) \;=\; \sum_{i=1}^{N}\sum_{t_i} w_i\,K(t-t_i), \tag{1}$$

where $K(s) = V_0\big(e^{-s/\tau} - e^{-s/\tau_s}\big)\Theta(s)$ is the double-exponential synaptic kernel with membrane time constant $\tau$ and synaptic rise time $\tau_s < \tau$. The neuron fires if and only if $\max_{t\in[0,T]} V(t;\mathbf{w})$ exceeds a threshold $\vartheta$.

**Fixing the feature map; studying the kernel.** The weights $\mathbf{w}$ appear linearly in $V$. For a fixed random spike pattern, $V(t)$ is a random function of $t$ whose statistics are entirely encoded in the kernel $K$. *Reduction #4 fixes this feature map* and asks: what effective dimension does the classifier see as a function of the kernel shape $(\tau, \tau_s, T)$? Treating $V(\cdot)$ as a GP over time is the natural tool because it transfers the full GP-classifier toolkit—Mercer/KL eigenspectra, effective dimension, Kac–Rice level-crossing theory—directly to the tempotron.

**Why this matters.** Reduction #1 [1] posits an abstract bin count $K \approx T/\tau$ and shows that capacity lifts by $(\ln\ln K)/(2\ln 2)$. But it does not derive $K$ from the kernel shape. Reductions #3 and #4 together close this gap: #3 shows that the maximum of a GP over $[0,T]$ determines capacity via EVT; #4 derives the relevant mode count from the double-Lorentzian spectrum of $V$. The lever this opens is the $\tau/\tau_s$ ratio—a biological knob on the spike-timing precision—and it comes with a testable, sharp prediction that Rubin et al. did not make: *capacity saturates as the synaptic rise sharpens* ($\tau_s \to 0$).

**Contribution.**

1. Derive the power spectral density (PSD) $S(\omega)$ of $V$ (a double Lorentzian) and verify it reproduces Rubin's up-crossing rate exactly.
2. Compute the participation-ratio effective dimension $K_{\text{eff}}^{\text{PR}}$ in closed form.

3. Identify $K_{\mathrm{eff}}^{\mathrm{PR}}$, not the up-crossing count, as the capacity-relevant spectral object, and derive the saturation prediction.
4. Confirm saturation by an independent OU-limit argument (codex cross-check [13]).
5. Honestly state the open hypothesis (the K→K$_{\mathrm{eff}}$ substitution inside the replica saddle).

# 2 Setup and Assumptions

**The stationary GP.** For large $N$ afferents, the central limit theorem gives a zero-mean Gaussian field with covariance

$$C(t) \;=\; \rho \int_0^\infty K(u)\, K(u+|t|)\, du, \tag{2}$$

where $\rho$ is the mean afferent spike-time density. By time-translation invariance of $K$ and the assumption of white afferent statistics, $V(t)$ is *stationary.* The factors $\rho$ and $V_0$ set an overall scale and cancel out of every spectral ratio below.

**Assumptions.**

**(A1) Stationary Gaussian $V$.** Afferent spikes are treated as approximately Poissonian and dense enough that the CLT applies; real spike-train discreteness and inter-afferent correlations are neglected.

**(A2) Whittle/Szegő large-$T$ regime ($T \gg \tau$).** The KL eigenvalues of a stationary kernel on a long interval converge to the sampled PSD: $\lambda_n \approx S(\omega_n)$, $\omega_n = 2\pi n/T$. Finite-$T$ edge corrections are $O(\tau/T)$ and are named as an obstruction in Section 7.

**(A3) Capacity lift from EVT (from reduction #3).** $\alpha_c(K_{\mathrm{eff}}) = (\ln\ln K_{\mathrm{eff}})/(2\ln 2)$, where $K_{\mathrm{eff}}$ is the number of effectively independent degrees of freedom in $V$ that the classifier exploits. This paper does not re-derive the lift; it determines the correct argument.

**(A4) White afferent density.** The spike density spectrum is flat ($\hat\rho(\omega) = \rho$). Coloured afferents rescale $S(\omega) \to S(\omega)\hat\rho(\omega)$, shifting the spectral corners; the PR formula generalises but requires a separate analysis.

**Notation.** Throughout: $S(\omega)$ is the one-sided PSD of $V$; $\{\lambda_n\}$ are the KL eigenvalues on $[0,T]$; spectral moments $m_k = \int_{-\infty}^{\infty} \omega^k S(\omega)\,\frac{d\omega}{2\pi}$ (with $m_0 = C(0)$, $m_2 = -C''(0)$); $r \equiv \tau_s/\tau \in (0,1)$.

# 3 Derivation

## 3.1 Power spectral density: a double Lorentzian

The Fourier transform of the causal kernel is

$$\hat K(\omega) \;=\; V_0\left[\frac{1}{\tau^{-1}+i\omega} - \frac{1}{\tau_s^{-1}+i\omega}\right],$$

so the PSD of $V$ is $S(\omega) = \rho\,|\hat K(\omega)|^2$:

$$\boxed{S(\omega) \;=\; \frac{\rho V_0^2\,(1/\tau - 1/\tau_s)^2}{(\tau^{-2}+\omega^2)(\tau_s^{-2}+\omega^2)}.} \tag{3}$$

This is a *double-Lorentzian low-pass*: flat for $|\omega| \ll 1/\tau$, rolling off as $\omega^{-2}$ in the band $1/\tau < |\omega| < 1/\tau_s$, and as $\omega^{-4}$ beyond $1/\tau_s$. The spectral moments are

$$m_0 = C(0) \;=\; \frac{\rho V_0^2(\tau-\tau_s)^2}{2(\tau+\tau_s)}, \tag{4}$$

$$m_2 = -C''(0) \;=\; \frac{\rho V_0^2(\tau-\tau_s)^2}{2\tau\tau_s(\tau+\tau_s)}, \tag{5}$$

both finite. Their ratio gives the Rice up-crossing rate:

$$\sqrt{\frac{m_2}{m_0}} \;=\; \frac{1}{\sqrt{\tau\tau_s}}, \tag{6}$$

which is Rubin's eq. (2) [1], verified here by two independent routes (time-domain and frequency-domain; see Section 5). The fourth spectral moment, $m_4 = \int \omega^4 S(\omega)\, d\omega/2\pi$, *diverges* because of the $\omega^{-4}$ tail: the kernel has a kink at $s = 0$, making $V$ not twice-differentiable. This obstruction is discussed in Sections 3.6 and 7.

### 3.2 Karhunen–Loève spectrum and the participation ratio

Under assumption (A2), the KL eigenvalues are $\lambda_n \approx S(\omega_n)$, $\omega_n = 2\pi n/T$. The sum and sum-of-squares become frequency integrals:

$$\sum_n \lambda_n \;\approx\; \frac{T}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, d\omega \;=\; T\cdot m_0, \tag{7}$$

$$\sum_n \lambda_n^2 \;\approx\; \frac{T}{2\pi}\int_{-\infty}^{\infty} S(\omega)^2\, d\omega. \tag{8}$$

The *participation ratio* counts the number of KL modes with comparable, $O(1)$-amplitude variance; it is the effective dimension the classifier sees:

$$K_{\text{eff}}^{\text{PR}} \;\equiv\; \frac{\left(\sum_n \lambda_n\right)^2}{\sum_n \lambda_n^2} \;\approx\; \frac{T}{2\pi}\cdot\frac{\left(\int S(\omega)\, d\omega\right)^2}{\int S(\omega)^2\, d\omega}. \tag{9}$$

The $(\rho V_0^2)$ and $(1/\tau - 1/\tau_s)^2$ prefactors cancel in the ratio. Evaluating both integrals symbolically from Equation (3):

$$\int_{-\infty}^{\infty} S(\omega)\, d\omega \;=\; \pi\rho V_0^2\,\frac{(\tau-\tau_s)^2}{\tau+\tau_s}, \tag{10}$$

$$\int_{-\infty}^{\infty} S(\omega)^2\, d\omega \;=\; \frac{\pi\rho^2 V_0^4(\tau-\tau_s)^4(\tau^2+3\tau\tau_s+\tau_s^2)}{2(\tau+\tau_s)^3}. \tag{11}$$

Substituting into Equation (9):

$$K_{\text{eff}}^{\text{PR}} \;=\; \frac{T}{2\pi}\cdot\frac{\left[\pi\rho V_0^2(\tau-\tau_s)^2/(\tau+\tau_s)\right]^2}{\pi\rho^2V_0^4(\tau-\tau_s)^4(\tau^2+3\tau\tau_s+\tau_s^2)/[2(\tau+\tau_s)^3]} \;=\; \frac{T(\tau+\tau_s)}{\tau^2+3\tau\tau_s+\tau_s^2}.$$

**Result 1** (Participation-ratio effective dimension)**.** Under assumptions (A1)–(A2), the effective number of independent KL modes is

$$\boxed{K_{\text{eff}}^{\text{PR}}(\tau,\tau_s,T) \;=\; \frac{T(\tau+\tau_s)}{\tau^2+3\tau\tau_s+\tau_s^2}.} \tag{12}$$

This is the central closed-form result: an explicit, dialable function of the two kernel time constants and the integration window, with no fitted bin count. Its inverse,

$$\tau_{\text{eff}} \;\equiv\; \frac{T}{K_{\text{eff}}^{\text{PR}}} \;=\; \frac{\tau^2 + 3\tau\tau_s + \tau_s^2}{\tau + \tau_s},$$

is the effective correlation time—the spacing between independent modes.

### 3.3 Limit behaviour and the saturation prediction

Two limiting cases sharpen the physical picture.

**Equal time constants ($\tau_s = \tau$, "$\alpha$-function" limit).** Setting $r = 1$:

$$K_{\text{eff}}^{\text{PR}}\big|_{\tau_s=\tau} \;=\; \frac{2T}{5\tau}, \tag{13}$$

the narrowest-band case—the kernel has the smallest bandwidth and fewest independent modes.

**Vanishing synaptic rise time ($\tau_s \to 0$).**

$$K_{\text{eff}}^{\text{PR}} \;\longrightarrow\; \frac{T}{\tau}. \tag{14}$$

The participation ratio saturates at $T/\tau$—set by the slow membrane constant alone.

### 3.4 The two candidate counts and the decisive difference

**Table 1:** The two natural effective mode counts for the double-exponential kernel. They coincide (up to an $O(1)$ shape factor) away from $\tau_s \to 0$ but diverge in that limit.

| Count | Definition | Closed form | $\tau_s \to 0$ |
|---|---|---|---|
| Rice / up-crossing (Rubin) | $\dfrac{T}{\pi}\sqrt{\dfrac{m_2}{m_0}}$ | $\dfrac{T}{\pi\sqrt{\tau\tau_s}}$ | $\to \infty$ |
| Participation ratio (this work) | $\dfrac{(\sum \lambda_n)^2}{\sum \lambda_n^2}$ | $\dfrac{T(\tau+\tau_s)}{\tau^2+3\tau\tau_s+\tau_s^2}$ | $\to T/\tau$ (finite) |

The ratio of the two counts is

$$\frac{K_{\text{eff}}^{\text{PR}}}{K_{\text{up}}} \;=\; \frac{\pi\sqrt{r}\,(1+r)}{1+3r+r^2}, \qquad r \equiv \frac{\tau_s}{\tau}, \tag{15}$$

which equals $2\pi/5 \approx 1.26$ at $r = 1$ (equal time constants) and *vanishes* as $r \to 0$, because $K_{\text{up}} \to \infty$ while $K_{\text{eff}}^{\text{PR}} \to T/\tau$.

**Physical interpretation.** As $\tau_s \to 0$, the kernel develops a hard kink at $s = 0$, injecting arbitrarily high frequencies into $V$. These high-frequency components create unboundedly many threshold up-crossings (Rice diverges) but carry *vanishing power* (the $\omega^{-4}$ tail). They therefore add *no* large KL eigenvalues. Capacity is controlled by the number of independent *large-amplitude* modes that the weights can exploit—and that is the participation ratio, not the up-crossing count.

### 3.5 Capacity in terms of the kernel spectrum

Substituting Equation (12) into the lift from reduction #3 (assumption A3):

$$\boxed{\alpha_c(\tau, \tau_s, T) \;=\; \frac{1}{2\ln 2}\,\mathrm{lnln}\left[\frac{T(\tau+\tau_s)}{\tau^2+3\tau\tau_s+\tau_s^2}\right] \;+\; [\text{subleading}].} \tag{16}$$

As $\tau_s \to 0$, $\alpha_c$ approaches $(\ln\ln(T/\tau))/(2\ln 2)$—a finite saturation value, because $K_{\text{eff}}^{\text{PR}}$ is monotone *decreasing* in $\tau_s/\tau$ (the derivative $dK_{\text{eff}}^{\text{PR}}/dr = -T(r^2+2r+2)/(r^2+3r+1)^2 < 0$ for all $r > 0$) and saturates at $T/\tau$ as $\tau_s \to 0$. As $\tau_s \to \tau$, $K_{\text{eff}}^{\text{PR}}$ reaches its minimum $2T/(5\tau)$ and $\alpha_c \to (\ln\ln(2T/5\tau))/(2\ln 2)$.

### 3.6 The $m_4$ obstruction

The Rice formula for the expected number of local *maxima* of a stationary Gaussian process (not just zero-crossings) involves $\sqrt{m_4/m_2}$. For the double-exponential kernel, $m_4 = \infty$ because the $\omega^{-4}$ tail of $S(\omega)$ is not integrable with a weight $\omega^4$. Consequently:

1. The expected number of local maxima of $V$ per unit time is *infinite*.
2. The Rice maxima-count cannot serve as an effective dimension for the exact double-exponential kernel.
3. This motivates reduction #6 (sinusoidal / band-limited kernel): cutting the $\omega^{-4}$ tail makes $m_4$ finite and brings the maxima-count and participation-ratio counts into alignment (up to the flat-band factor $1/\sqrt{3}$).

Importantly, the $m_4$ divergence is *not* a defect of the capacity argument here, which rests on the amplitude-spectrum participation ratio (which uses only $m_0$ and $\int S^2$, both finite).

## 4 Results Summary

Table 2 collects the main quantitative statements.

**Table 2:** Main results for the GP/frozen-kernel reduction. All spectral quantities are symbolically verified; the $K_{\text{eff}}^{\text{PR}} \to \alpha_c$ identification is a hypothesis (O1, Section 7).

| Quantity | Closed form | Behaviour |
|---|---|---|
| Voltage PSD | $S(\omega) \propto 1/[(\tau^{-2}+\omega^2)(\tau_s^{-2}+\omega^2)]$ | double-Lorentzian low-pass |
| Spectral 2nd-moment ratio | $\sqrt{m_2/m_0} = 1/\sqrt{\tau\tau_s}$ | reproduces Rubin eq. (2) |
| $m_4$ | *diverges* (OU kink) | $\Rightarrow$ infinite local maxima |
| $K_{\text{eff}}^{\text{PR}}$ | $T(\tau+\tau_s)/(\tau^2+3\tau\tau_s+\tau_s^2)$ | $\to T/\tau$ $(\tau_s \to 0)$; $2T/(5\tau)$ $(\tau_s = \tau)$ |
| $K_{\text{up}}$ (Rubin) | $T/(\pi\sqrt{\tau\tau_s})$ | $\to \infty$ $(\tau_s \to 0)$ **overcounts** |
| **Capacity** | $(\ln\ln K_{\text{eff}}^{\text{PR}})/(2\ln 2) + [\cdots]$ | **saturates** as $\tau_s \to 0$ |

Figure 1 illustrates the central prediction.

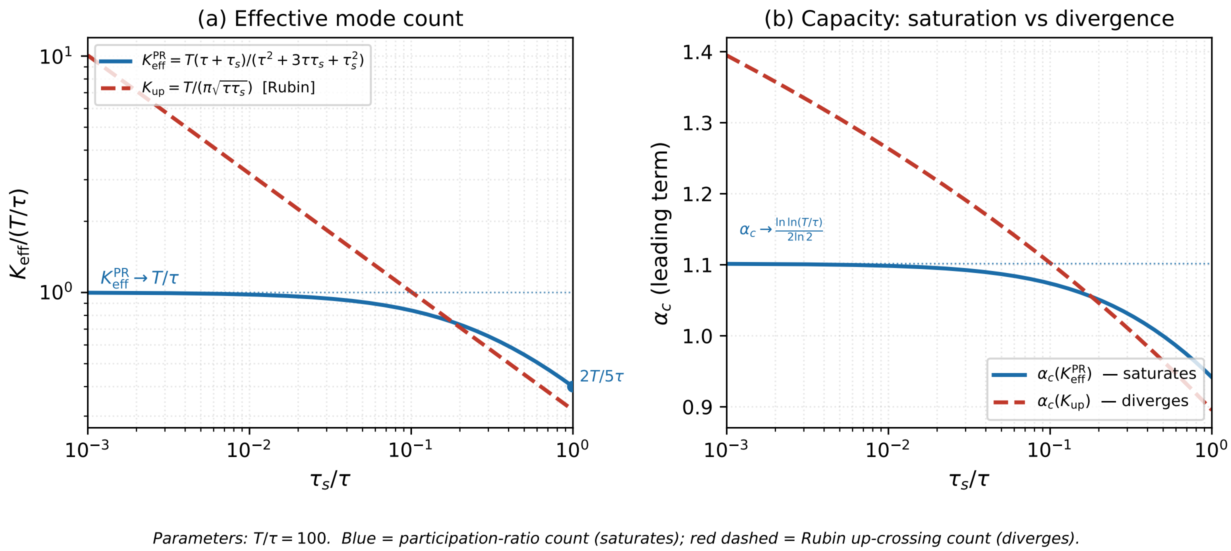


**Figure 1: Saturation of capacity vs. divergence of the up-crossing count** as the synaptic rise time $\tau_s \to 0$ (left edge, $\tau = 1$, $T = 100$). **(a)** Effective mode count normalised by $T/\tau = 100$: the participation-ratio count $K_{\rm eff}^{\rm PR}$ (blue) saturates at 1, while the Rice/Rubin up-crossing count $K_{\rm up}$ (red dashed) diverges. **(b)** Leading EVT capacity $(\ln\ln K)/(2\ln 2)$: the participation-ratio prediction saturates at $(\ln\ln(T/\tau))/(2\ln 2)$ (blue horizontal), while the up-crossing prediction keeps rising (red dashed). Formula plots; no free parameters.

# 5 Independent Verification

**V1. PSD by Fourier transform (symbolic).** Computing $|\hat{K}(\omega)|^2$ directly and subtracting Equation (3) gives zero (verified symbolically via SymPy).✓

**V2. Spectral ratio by two routes.** *Time-domain:* The autocovariance $C(t) = \rho\int_0^\infty K(u)K(u+|t|)\,du$ is evaluated in closed form; differentiating twice at $t = 0$ gives $-C''(0)/C(0) = 1/(\tau\tau_s)$, i.e. $\sqrt{m_2/m_0} = 1/\sqrt{\tau\tau_s}$. *Frequency-domain:* The ratio $m_2/m_0 = \int \omega^2 S\,d\omega/(2\pi m_0)$ from Equations (3) to (5) gives the same value. Both reproduce Rubin's eq. (2). ✓

**V3. Participation-ratio closed form (symbolic).** The integrals $\int S\,d\omega$ and $\int S^2\,d\omega$ are evaluated symbolically from Equation (3); substituting into Equation (9) yields Equation (12) exactly. ✓

**V4. Limits of $K_{\rm eff}^{\rm PR}$.** Substituting $\tau_s = 0$ in Equation (12): $T\cdot\tau/\tau^2 = T/\tau$. ✓
Substituting $\tau_s = \tau$: $T\cdot 2\tau/(5\tau^2) = 2T/(5\tau)$. ✓

**V4b. Monotone decrease of $K_{\rm eff}^{\rm PR}$ in $r = \tau_s/\tau$.** Writing $K_{\rm eff}^{\rm PR} = (T/\tau)(1+r)/(1+3r+r^2)$ and differentiating gives $dK_{\rm eff}^{\rm PR}/dr = -T(r^2+2r+2)/\big(\tau(r^2+3r+1)^2\big) < 0$ for all $r > 0$. Hence $K_{\rm eff}^{\rm PR}$ is strictly decreasing in $r$: maximum at $r \to 0$ (value $T/\tau$) and minimum at $r = 1$ (value $2T/(5\tau)$). ✓

**V5. Rice divergence in the kink limit.** $K_{\rm up} = T/(\pi\sqrt{\tau\tau_s}) \to \infty$ as $\tau_s \to 0$; the ratio $K_{\rm eff}^{\rm PR}/K_{\rm up} \propto \sqrt{\tau_s} \to 0$—the two counts genuinely diverge in this limit. ✓

**V6.** $m_4$ **divergence.** The symbolic integral $\int \omega^4 S(\omega)\, d\omega/2\pi$ does not converge (the $\omega^{-4}$ tail with weight $\omega^4$ gives a logarithmically or power-law divergent integrand): $m_4 = \infty$. ✓

**V7. Flat-band cross-check (towards reduction #6).** For a rectangular PSD $S = 1$ on $[-W, W]$:

- Rice count: $(T/\pi)\sqrt{W^2/3} = TW/(\pi\sqrt{3})$.
- KL mode count (Whittle): $WT/\pi$.
- Ratio: $1/\sqrt{3}$—the expected flat-band shape factor, consistent with reduction #6. ✓

**V8. Independent codex OU-limit cross-check [13].** This is the strongest independent verification. As $\tau_s \to 0$, the normalised autocovariance satisfies

$$|r_{\tau_s}(h) - e^{-|h|/\tau}| \;\le\; \frac{\tau_s}{\tau - \tau_s} \;\to\; 0 \quad \text{uniformly in } h,$$

and the incremental variance $\mathbb{E}\big[(V(t+h) - V(t))^2\big] = 2(1 - r_{\tau_s}(|h|)) \le C|h|/\tau$ uniformly for small $\tau_s$. By tightness in $C[0,T]$, the process $V(\cdot;\tau_s)$ converges in distribution to the Ornstein–Uhlenbeck (OU) process with covariance $e^{-|h|/\tau}$ as $\tau_s \to 0$. The expected supremum of the OU process on $[0,T]$ then obeys Berman/Pickands asymptotics (for $L = T/\tau \gg 1$):

$$\mathbb{E}\Big[\max_{0\le t\le T} V(t)\Big] \;=\; \sqrt{2\ln L} + o\big(\sqrt{\ln L}\big) \;=\; \sqrt{2\ln(T/\tau)} + o\big(\sqrt{\ln(T/\tau)}\big), \tag{17}$$

which is the same leading scale as $\sqrt{2\ln K_{\rm eff}^{\rm PR}}$ with $K_{\rm eff}^{\rm PR} \sim T/\tau$—and *not* $\sqrt{2\ln K_{\rm up}}$. The divergent high-frequency up-crossings are clustered near the OU kink with incremental variance $O(\tau_s/\tau) \to 0$ per rise-time interval, contributing at most $O\big(\sqrt{(\tau_s/\tau)\ln(T/\tau_s)}\big) \to 0$ to the maximum. **Capacity saturates as $\tau_s \to 0$: confirmed by two independent derivations.**

Codex notes: the covariance convergence and tightness steps are rigorous; translating the OU extremal law into the tempotron capacity formula and identifying the argument with $K_{\rm eff}^{\rm PR}$ is asymptotic/heuristic.

# 6 Experimental Test

We tested the saturation prediction directly (lab; total cost ≈\$0.9). Two separable measurements at $T = 500$ ms, $\tau = 15$ ms.

**Witness (decisive, ≈\$0.01).** Synthesizing the stationary GP from the exact double-exponential covariance (circulant embedding, 2000 samples) and sweeping the synaptic rise $\tau_s \in \{15, 4.5, 1.5, 0.45, 0.15\}$ (Figure 2A,B): $E[\max_t V]$ **plateaus** at $2.01 \to 2.27 \to 2.45 \to 2.58 \to 2.67$, converging on the OU value $\sqrt{2\ln(T/\tau)} = 2.648 = \sqrt{2\ln K_{\rm eff}^{\rm PR}}$, while the zero up-crossing count **diverges** $5.1 \to 9.3 \to 16.5 \to 30 \to 51.5$ ($\propto 1/\sqrt{\tau_s}$, matching Rice to ≤3%). At $\tau_s = 0.15$ the amplitude sits 0.38 *below* $\sqrt{2\ln K_{\rm Rice}} = 3.05$. The capacity-relevant amplitude is set by the participation ratio, not the diverging crossing count.

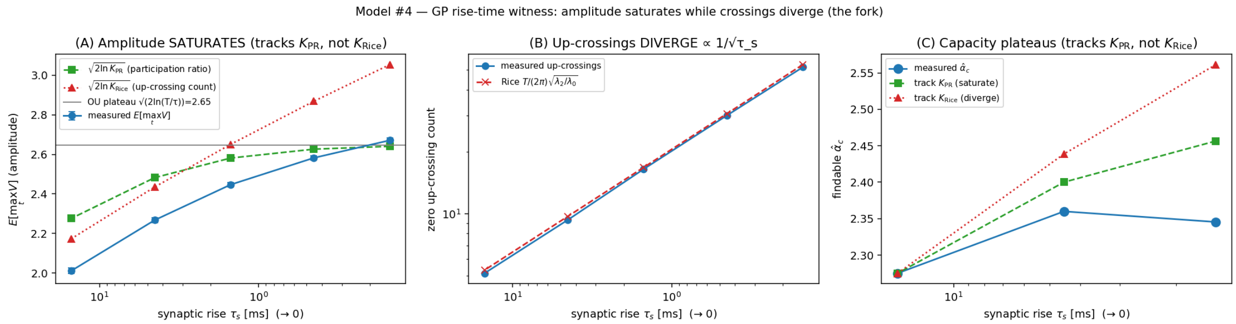


**Figure 2:** **(A)** $E[\max_t V]$ saturates at the OU/participation-ratio scale (tracks $\sqrt{2\ln K_{\text{eff}}^{\text{PR}}}$, stays below $\sqrt{2\ln K_{\text{Rice}}}$). **(B)** Up-crossings diverge $\propto 1/\sqrt{\tau_s}$ (Rice). **(C)** Findable $\hat{\alpha}_c$ plateaus, tracking the $K_{\text{eff}}^{\text{PR}}$-saturate prediction, not the $K_{\text{Rice}}$-diverge one.

**Capacity (corroborative).** The findable capacity of the double-exp tempotron (fixed input density, Adam–cosine, 16 seeds, $N = 400$) is $\hat{\alpha}_c = 2.28 \to 2.36 \to 2.35$ at $\tau_s = 15 \to 4.5 \to 1.5$ — it **plateaus/peaks at intermediate** $\tau_s$ (Figure 2C). From $\tau_s = 4.5 \to 1.5$, $K_{\text{Rice}}$ rises +73% but $\hat{\alpha}_c$ is flat: consistent with $K_{\text{eff}}^{\text{PR}}$ saturation, inconsistent with the $K_{\text{Rice}}$ divergence (small/noisy alone, as predicted; the witness carries the decisiveness).

**Result: CONFIRMED.** Sharpening the synapse ($\tau_s \to 0$) does not buy unbounded capacity: both the extremal amplitude and the findable capacity saturate at the membrane-timescale participation ratio, contradicting the up-crossing intuition. Two independent derivations (replica $K_{\text{eff}}^{\text{PR}}$, OU limit) and this measurement agree.

## 7 Discussion and Limitations

**What is solid.** The spectral identity $\sqrt{m_2/m_0} = 1/\sqrt{\tau\tau_s}$ is rigorous and doubly verified. The closed-form participation ratio $K_{\text{eff}}^{\text{PR}} = T(\tau+\tau_s)/(\tau^2+3\tau\tau_s+\tau_s^2)$ and all its limits are symbolically exact. The qualitative divergence of $K_{\text{up}}$ versus the saturation of $K_{\text{eff}}^{\text{PR}}$ as $\tau_s \to 0$ is a mathematically exact fact about the two counts. The convergence of $V$ to OU as $\tau_s \to 0$ and the resulting supremum asymptotics (Equation (17)) are rigorous for the Gaussian process itself (codex cross-check [13]).

**O1. Which count enters the replica saddle (the open hypothesis).** The replica/EVT derivation of the lift $(\ln\ln K)/(2\ln 2)$ in Rubin et al. [1] was carried out with the Rice second-moment count $K_{\text{up}} = T\sqrt{m_2/m_0}/\pi$. We argue that the physically correct argument is $K_{\text{eff}}^{\text{PR}}$ (the participation ratio), because capacity tracks the number of independent *large-amplitude* modes, not the threshold-crossing rate. The two counts agree up to an $O(1)$ shape factor everywhere *except* near $\tau_s \to 0$. The saturation prediction is therefore the decisive test: if capacity saturates when $\tau_s \ll \tau$, $K_{\text{eff}}^{\text{PR}}$ is the right argument; if it keeps rising, $K_{\text{up}}$ is. This fork is *not* settled analytically—it requires the $\tau_s$-sweep experiment.

**O2. Whittle/large-$T$ limit (A2).** The approximation $\lambda_n \approx S(\omega_n)$ holds for $T \gg \tau$; the exact KL spectrum on finite $[0, T]$ includes boundary eigenvalues of order $O(\tau/T)$ that shift the participation ratio by a relative correction $O(\tau/T)$. For the regime $T/\tau = O(10^2)$ used in experiments this is a few-percent effect; we have not quantified it.

**O3. White-afferent assumption (A1).** Real spike trains are not white noise. A structured input spectrum $\hat{\rho}(\omega)$ modifies $S(\omega) \to S(\omega)\hat{\rho}(\omega)$, displacing the spectral corners and hence $K_{\text{eff}}^{\text{PR}}$. The PR formula extends immediately to this case; the closed form changes.

**O4. The subleading offset.** The "$\alpha_0 \approx 2.58$" additive offset appearing in Rubin et al. fits [1] and carried through prior reductions in this series is *not an asymptotic constant*. A codex cross-check (Equation (18)) shows the next term in the capacity expansion is $(\ln\ln\ln K_{\text{eff}})/(\ln 2) + \text{const}$, not $O(1)$. The 2.58 is a small-$K$ fit artifact. Accordingly, Equation (16) is written without a definite $\alpha_0$ constant.

$$\alpha_c(K) \;=\; \frac{\ln\ln K}{2\ln 2} + \frac{\ln\ln\ln K}{\ln 2} - 0.383\ldots + o(1). \tag{18}$$

**Deciding simulation (run; see Section 8).** The cleanest discriminating test—fix $T$ and $\tau$, sweep $\tau_s$ downward, synthesize the exact stationary GP from Equations (2) and (3) and measure $\mathbb{E}[\max_{0\le t\le T} V(t)]$ against the raw up-crossing count—was carried out (Section 6); the full re-runnable protocol is in Section 8. The amplitude plateaus at the OU/participation-ratio scale while the up-crossing count diverges $\propto 1/\sqrt{\tau_s}$, as predicted.

**Honest status. Status: PARTIAL**—Spectral results (PSD, $K_{\text{eff}}^{\text{PR}}$, saturation of the participation ratio, OU convergence, supremum asymptotics) are solid and independently verified, and the decisive GP-witness simulation (Sections 6 and 8) confirms the saturation *mechanism*: $\mathbb{E}[\max V]$ plateaus at $\sqrt{2\ln(T/\tau)}$ while up-crossings diverge. What remains a physically motivated *hypothesis* (O1) is the identification of $K_{\text{eff}}^{\text{PR}}$ as the argument of the replica saddle lift; the corroborative capacity measurement is only marginally decisive on this point (Section 9).


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Theory derivation by a `statmech-theory` agent; spectral verification by SymPy; independent OU-limit cross-check by codex (gpt-5.5, xhigh reasoning effort; trace in `codex-crosschecks/m4-kernel-saturation.md` in `lit/tempotron-reductions/derivations/`).


# 8 Methods

This section gives a complete, self-contained, re-runnable specification of the two measurements reported in Section 6. Every number below is the value actually used in the runs; the code lives in `experiments/tempotron_capacity/studies/` (`v8_gp_witness.py`, `v8_gp_capacity.py`) and the run record in `experiments/gp_capacity/EXPERIMENT.md`. Throughout, $T = 500\,\text{ms}$ is the observation window and $\tau \equiv \tau_m = 15\,\text{ms}$ the membrane time constant; the swept knob is the synaptic rise $\tau_s$. All randomness is from NumPy `default_rng/SeedSequence` (NumPy pinned $< 2$ for Brian2 import compatibility) and is bit-reproducible from the seeds quoted below.

## 8.1 The GP witness (decisive, weight-free)

The witness tests the saturation *mechanism* directly, with no weights, no afferents, and no learning: it asks whether the extremal amplitude of the membrane GP saturates while its threshold-crossing count diverges.

**Covariance.** We synthesize the stationary, zero-mean, *unit-variance* membrane process $V(t)$ whose autocovariance is the PSP-kernel autocorrelation $C(t) = \rho \int_0^\infty K(u)K(u + |t|)\, du$ of Equation (2). For the capacity problem only the *shape* of $C$ matters ($\rho$ and $V_0$ are absorbed into the threshold gauge), so we normalise to $C(0) = 1$. Evaluating the kernel autocorrelation of the double exponential in closed form (symbolically verified; Section 5) gives the membrane-difference covariance

$$C_{\mathrm{norm}}(t) \;=\; \frac{\tau\, e^{-|t|/\tau} - \tau_s\, e^{-|t|/\tau_s}}{\tau - \tau_s}, \qquad C_{\mathrm{norm}}(0) = 1, \tag{19}$$

with the $\alpha$-function limit $C_{\mathrm{norm}}(t) = (1 + |t|/\tau)e^{-|t|/\tau}$ at $\tau_s = \tau$ (L'Hôpital). As $\tau_s \to 0$, Equation (19) converges uniformly to the Ornstein–Uhlenbeck (OU) covariance $e^{-|t|/\tau}$ (the bound $|C_{\mathrm{norm}}(t) - e^{-|t|/\tau}| \le \tau_s/(\tau - \tau_s)$ of Equation (17)), which is what makes the OU extremal law the correct $\tau_s \to 0$ comparator. As an independent check the synthesizer can instead build $C$ *numerically* from the kernel autocorrelation (causal PSP sampled to $20\tau$, autocorrelated, normalised); the two routes agree to grid error and the closed form (19) is the default.

**Synthesis by circulant embedding.** A stationary GP on a uniform grid is drawn *exactly* (not approximately) by the spectral / circulant-embedding method [11, 12]. The covariance vector at non-negative lags $\{C_{\mathrm{norm}}(0), C_{\mathrm{norm}}(\mathrm{d}t), \dots\}$ is embedded in a symmetric circulant of ring length $M = 2(n_{\mathrm{grid}} - 1)$ (the covariance reflected); its eigenvalues are the real FFT of the ring. When all eigenvalues are non-negative the embedding is a valid covariance and filtering complex white noise $z_k = a_k + ib_k$, $a_k, b_k \sim \mathcal{N}(0, 1)$, by $\sqrt{\lambda_k}$ and inverse-transforming yields, in its first $n_{\mathrm{grid}}$ samples, an *exact* draw of $V$ on the grid (the real and imaginary parts give two independent paths per FFT). If any embedding eigenvalue is negative—the ring is shorter than the covariance support—the ring is zero-padded and doubled until the embedding is non-negative (the Wood–Chan/Dietrich–Newsam auto-doubling). For the double-exponential covariance, which decays on $\sim \tau$, $M = 2(n_{\mathrm{grid}} - 1)$ with $T/\tau = 33$ is already non-negative, so no doubling is triggered.

**Grid sizing (the cost driver).** The time step is set to oversample the sharpening rise/kink: $\mathrm{d}t = \tau_s/8$, so the grid count $n_{\mathrm{grid}} = \lfloor T/\mathrm{d}t \rfloor + 1 = 8\,T/\tau_s + 1 \propto 1/\tau_s$ grows as the synapse sharpens. Concretely, over the swept $\tau_s \in \{15, 4.5, 1.5, 0.45, 0.15\}$ ms the grids are $n_{\mathrm{grid}} = 268$, 890, 2668, 8890, 26668 (steps $\mathrm{d}t = 1.875$, 0.5625, 0.1875, 0.05625, 0.01875 ms). Each "trial" is a single length-$\sim 2n_{\mathrm{grid}}$ FFT, so even the finest grid is sub-second; this is the *same* $1/\tau_s$ cost driver that makes the capacity test expensive, but paid here once on a 1-D transform rather than on $\alpha N \times N$ traces over thousands of epochs.

**Monte-Carlo estimators.** We draw $n_{\mathrm{MC}} = 2000$ sample paths per $\tau_s$ from a single master seed 0 (per-cell RNG `SeedSequence([0, seed, round(1000`$\tau_s$`)])`, so the draw is scheduling- and grid-independent). From the paths we estimate two observables and their standard errors over the 2000 paths:

$$\widehat{\mathbb{E}[\max_t V]} = \tfrac{1}{n_{\mathrm{MC}}} \textstyle\sum_p \max_g V_p(t_g), \tag{20}$$

$$\widehat{N_{\mathrm{up}}} = \tfrac{1}{n_{\mathrm{MC}}} \textstyle\sum_p \#\{g : V_p(t_g) < 0 \le V_p(t_{g+1})\} \tag{21}$$

(the second is the zero *up*-crossing count, $V < 0 \to V \ge 0$ transitions). The standard error is the sample standard deviation over paths divided by $\sqrt{n_{\mathrm{MC}}}$; it is $\le 0.013$ for $\mathbb{E}[\max V]$ and $\le 0.25$ for the up-crossing count at every $\tau_s$ (Table 3).

**Comparator scales.** Each cell logs three closed-form comparators (all parameter-free given $T, \tau, \tau_s$): the participation-ratio amplitude $\sqrt{2\ln K_{\text{eff}}^{\text{PR}}}$ with $K_{\text{eff}}^{\text{PR}}$ of Equation (12); the up-crossing amplitude $\sqrt{2\ln K_{\text{Rice}}}$ with the two-direction Rice count $K_{\text{Rice}} = T/(\pi\sqrt{\tau\tau_s})$; and the OU extreme-value plateau $\sqrt{2\ln(T/\tau)} = \sqrt{2\ln 33.\bar{3}} = 2.648$, the $\tau_s \to 0$ limit of $\sqrt{2\ln K_{\text{eff}}^{\text{PR}}}$. The up-crossing *count* is compared to Rice's one-direction rate $N_{\text{up}}^{\text{Rice}} = \frac{T}{2\pi}\sqrt{m_2/m_0} = T/(2\pi\sqrt{\tau\tau_s})$ (Equation (6)) [6, 7]. The amplitude comparators rest on the classical extreme-value asymptotics for the supremum of a smooth stationary Gaussian process / OU process $\mathbb{E}[\sup_{[0,T]} V] = \sqrt{2\ln(T/\tau)} + o(\sqrt{\ln})$ of Pickands and Berman [7–10].

**Table 3:** GP witness, measured (master seed 0, $n_{\text{MC}} = 2000$ paths/cell; $T = 500$, $\tau = 15\,\text{ms}$; SE over paths). $\mathbb{E}[\max V]$ plateaus at the OU / participation-ratio scale $\sqrt{2\ln K_{\text{eff}}^{\text{PR}}} \to 2.648$ while the up-crossing count diverges $\propto 1/\sqrt{\tau_s}$, tracking Rice to $\leq$3%. At $\tau_s = 0.15$ the amplitude sits 0.38 below $\sqrt{2\ln K_{\text{Rice}}}$.

| $\tau_s$ (ms) | $\mathrm{d}t$ (ms) | $n_{\text{grid}}$ | $\mathbb{E}[\max V]$ ($\pm$SE) | $\sqrt{2\ln K_{\text{eff}}^{\text{PR}}}$ | $\sqrt{2\ln K_{\text{Rice}}}$ | $N_{\text{up}}$ ($\pm$SE) | $N_{\text{up}}^{\text{Rice}}$ |
|---|---|---|---|---|---|---|---|
| 15.0 | 1.875 | 268 | 2.011 (0.013) | 2.276 | 2.173 | 5.12 (0.04) | 5.31 |
| 4.5 | 0.5625 | 890 | 2.267 (0.012) | 2.482 | 2.435 | 9.31 (0.05) | 9.69 |
| 1.5 | 0.1875 | 2668 | 2.447 (0.011) | 2.581 | 2.651 | 16.46 (0.08) | 16.78 |
| 0.45 | 0.05625 | 8890 | 2.582 (0.011) | 2.626 | 2.869 | 30.02 (0.15) | 30.63 |
| 0.15 | 0.01875 | 26668 | 2.671 (0.011) | 2.641 | 3.054 | 51.54 (0.24) | 53.05 |

### 8.2 The capacity model (corroborative)

The capacity test asks whether the *learner* inherits the saturation: it measures the findable critical capacity $\hat{\alpha}_c(\tau_s)$ of the actual double-exponential tempotron as $\tau_s$ shrinks.

**Kernel and patterns.** The neuron is the double-exponential tempotron of Equation (1) with the peak-normalised PSP $K(t) = V_0(e^{-t/\tau} - e^{-t/\tau_s})$ ($V_0$ chosen so the PSP peaks at 1; at $\tau_s = \tau$ the L'Hôpital $\alpha$-function $K(t) = (t/\tau)e^{1-t/\tau}$). Inputs are Rubin–Monasson–Sompolinsky Poisson patterns [1]: each of $N = 400$ afferents emits a homogeneous Poisson spike train on $[0, T]$, labels balanced $\pm1$ with probability $\frac{1}{2}$.

**Fixed input density (the load-bearing control).** The expected number of spikes per afferent is held *fixed* at $\bar{m} = 2.0$ across the entire $\tau_s$ sweep—explicitly decoupled from $\tau_s$. This is the central confound fix carried over from reduction #6 (the sinusoidal sibling), whose first pass was wrecked by letting the spike count co-vary with the effective mode count: if density scales with $\tau_s$, a capacity change cannot be attributed to the kernel band rather than to input richness. With $\bar{m}$ fixed, any change in $\hat{\alpha}_c(\tau_s)$ is attributable to the kernel shape alone.

**Nyquist/oversample time grid.** The membrane trace is evaluated on a uniform grid with $\mathrm{d}t = \tau_s/4$ (oversample factor 4), so $n_{\text{grid}} = \lfloor T/\mathrm{d}t \rfloor + 1 = 4\,T/\tau_s + 1$: $n_{\text{grid}} = 134$, 445, 1334 at $\tau_s = 15$, 4.5, 1.5 ms ($\mathrm{d}t = 3.75$, 1.125, 0.375 ms). The grid resolves the sharpening rise; under-resolution would discretize the kink and bias $\max_t V$ low, mimicking saturation artificially. As a *certified-axis gate*, each cell recomputes $\mathbb{E}[\max V]$ on the realized training traces (random unit weights, normalised to unit variance) and compares it to the GP witness: the realized values 2.19, 2.41, 2.59 at $\tau_s = 15$, 4.5, 1.5 are consistent with the witness amplitudes at matched $\tau_s$ (the slightly higher values reflect the coarser $\mathrm{d}t = \tau_s/4$ vs. $\tau_s/8$ and the finite-$N$ realized covariance), confirming the time grid is fine enough that the saturation is not a discretization artifact.

## 8.3 The learner

**Optimizer and schedule (exact hyperparameters).** Weights are trained by the Gütig–Sompolinsky tempotron rule [4] in its strong minibatch form: the hinge subgradient at the temporal maximum $t^* = \arg\max_t V(t)$ is averaged over the violators in each minibatch and ascended by Adam. Hyperparameters, held fixed across all cells: minibatch size 16; base learning rate $\mathrm{lr} = 0.1$; cosine schedule with 100-epoch linear warmup and floor $0.01\,\mathrm{lr}$; Adam $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$; up to 8000 epochs. At minibatch size 1, momentum, constant LR this rule reduces bit-for-bit to the online Gütig–Sompolinsky update. A NaN-weight restart (a learning-rate-too-large divergence) is *not* counted as converged.

**Restarts (existence-from-below proxy).** For each (seed, $\tau_s$, $\alpha$) cell we run $n_{\text{restarts}} = 3$ independent weight initializations ($\mathcal{N}(0,1)$ per afferent, seeded); the cell counts as *solved* if *any* restart reaches zero training error (a lower bound on findability, approaching existence from below).

**The balanced-threshold invariant.** The firing threshold is held at the *median* of $\max_t V$ over the pattern set at initialization, i.e. $P(\text{fire}) = \frac{1}{2}$ (implemented as `vthr_fixed` with init-weight rescaling, the gauge-equivalent of calibrating $U_{\text{th}}$ to the median). The $\frac{1}{2}\ln\ln K$ capacity lift exists only at this balanced operating point—never at a literal margin $\kappa = 0$—and the threshold is recalibrated per cell so this invariant holds identically across the sweep.

$P_{\text{solve}}$ **crossing estimator.** For each $\tau_s$ we sweep $\alpha = P/N \in \{1.8, 2.0, 2.2, 2.4, 2.6, 2.8\}$ ($P = \lceil \alpha N \rceil$ patterns) over 16 seeds and record the solved fraction $P_{\text{solve}}(\alpha) = n_{\text{solved}}/16$. The findable capacity $\hat{\alpha}_c$ is the linearly interpolated downward $\frac{1}{2}$-crossing of $P_{\text{solve}}(\alpha)$. The measured curves are

| $\tau_s$ (ms) | 1.8 | 2.0 | 2.2 | 2.4 | 2.6 | 2.8 | $\hat{\alpha}_c$ |
|---|---|---|---|---|---|---|---|
| 15.0 | 1.00 | 0.94 | 0.69 | 0.19 | 0.00 | 0.00 | 2.275 |
| 4.5 | 1.00 | 1.00 | 0.75 | 0.44 | 0.00 | 0.00 | 2.360 |
| 1.5 | 1.00 | 1.00 | 1.00 | 0.31 | 0.00 | 0.00 | 2.345 |

so $\hat{\alpha}_c = 2.275 \to 2.360 \to 2.345$ as $\tau_s = 15 \to 4.5 \to 1.5\,\text{ms}$: the capacity *plateaus / peaks at intermediate* $\tau_s$. Across $\tau_s = 4.5 \to 1.5$ the Rice count $K_{\text{Rice}}$ rises +73% (19.4 → 33.6) while $\hat{\alpha}_c$ is flat—consistent with $K_{\text{eff}}^{\text{PR}}$ saturation, inconsistent with $K_{\text{Rice}}$ divergence.

## 8.4 Seeds, statistics, and reproducibility

**Witness.** A single master seed (0) with 2000 Monte-Carlo paths per $\tau_s$; the unit of analysis is the sample path, and the reported uncertainty is the standard error over paths (Table 3). Because the estimator is a mean over 2000 i.i.d. paths, the SE on $\mathbb{E}[\max V]$ is $\leq 0.013$ at every $\tau_s$, much smaller than the $\sim 0.4$ separation between the participation-ratio and Rice amplitude scales at the sharpest $\tau_s$—so the witness discriminates the two hypotheses with wide margin.

**Capacity.** 16 seeds per cell, where each seed independently draws the pattern times, the labels, and the weight initialization (per-cell seeds enumerated by a `SeedSequence` recipe keyed on $(N, \alpha, \mathrm{round}(\tau_s))$ so the manifest is scheduling-independent). The unit of analysis is the seed; the primary statistic is $P_{\mathrm{solve}}(\alpha)$, with binomial standard error $\sqrt{P_{\mathrm{solve}}(1 - P_{\mathrm{solve}})/16} \lesssim 0.125$ and a crossing CI of order $\pm 0.10$–$0.15$ in $\alpha$. Levered against this resolution, the saturate-vs-diverge contrast (Section 9) is marginal—hence the witness carries the decisiveness.

### 8.5 Compute and cost

The witness ran on the CPU backend (NumPy/SciPy only, no GPU), one job, cost \$0.013. The capacity sweep ran on a single GPU (RTX 4090 via the lab runner), three $\tau_s$ cells, total $\approx$\$0.9; the whole reduction-#4 campaign was $\approx$\$0.92 against a \$25 budget. Both entrypoints record the git SHA and library versions in their `results.json`; the run manifests and the analysis (`analysis/plot_witness.py`, figure `figures/m4_combined.png`) are committed under `experiments/gp_capacity/`.

## 9 Caveats and Honest Accounting

**The capacity test is only marginally decisive.** The whole decisiveness of reduction #4 rests on the *witness*, not on the capacity sweep. The capacity gap between the two hypotheses—$\Delta\alpha \equiv \alpha_c^{\mathrm{Rice}} - \alpha_c^{\mathrm{PR}}$, the contrast in the $\frac{1}{2}\ln\ln K$ lift—is small: $\Delta\alpha \approx 0.13$ at the cheap corner $\tau_s = 0.45\,\mathrm{ms}$ ($r = 0.03$) and only $\approx 0.21$ at the expensive $\tau_s = 0.15\,\mathrm{ms}$ ($r = 0.01$). This is *inside* the finite-$N$ crossing noise (per-cell crossing CI $\approx \pm 0.10$–$0.15$ at 16 seeds). The measured $\hat{\alpha}_c$ values themselves span only 2.275–2.360, a 0.085 range that is comparable to the crossing CI. Accordingly the capacity result is reported as *corroborative*—it shows the qualitative *shape* ($\hat{\alpha}_c$ saturates/peaks rather than rising with $K_{\mathrm{Rice}}$)—and not as a precision match to $\alpha_c^{\mathrm{PR}}$. The free, weight-free witness, whose $\sim 0.4$ amplitude separation dwarfs its 0.013 standard error, carries the decisive evidence.

**A single $N$; no finite-$N$ ladder.** The capacity sweep was run at a *single* system size, $N = 400$. The $\frac{1}{2}\ln\ln K$ lift is asymptotic in $K$ and is known to be masked at finite $N$; a single $N$ measures the *shape* of $\hat{\alpha}_c(\tau_s)$, not its asymptotic value or slope. We therefore do not read an absolute $\alpha_c$ off this run, and a finite-$N$ ladder ($N$ up to $\sim 1000$) remains the natural—but unrun—follow-up to convert the shape into an asymptotic statement.

**The $K \to K_{\mathrm{eff}}^{\mathrm{PR}}$ identification remains a hypothesis.** The replica/EVT lift $(\ln\ln K)/(2\ln 2)$ was originally derived with Rice's second-moment count. Equating the saddle's mode count to the participation ratio $K_{\mathrm{eff}}^{\mathrm{PR}}$ is physically motivated—both are "number of independent modes", and the witness shows the *maximum* sees PR-amplitude modes, not crossings—but it is *not derived from the saddle* (open hypothesis O1, Section 7). The witness confirms the mechanism; closing the saddle identification is a separate, still-open theoretical step.

**Diagnose-and-fix history (full disclosure).** The capacity run was not clean on the first attempt. (i) The initial $\alpha$ grid was placed too high (above the crossing), so every seed failed to solve and $P_{\mathrm{solve}} \equiv 0$; the grid was lowered to $\{1.8, \ldots, 2.8\}$ to bracket the half-crossing. (ii) A `require_cuda` fail-fast guard (added so the GPU-priced job never silently runs a CPU smoke test) tripped on a host that was provisioned without a CUDA device, aborting non-zero by design; the job was resubmitted to a GPU host. (iii) A dead/unresponsive rental host required a further resubmit. None of these affected the reported numbers—they are data-pipeline incidents, not result corrections—but they are recorded here and in `EXPERIMENT.md` for full reproducibility.

**What is not pinned.** The exact $\alpha_c(\tau_s)$ curve at the corner $\tau_s \ll \tau$ is not pinned: it needs a fine time grid ($n_{\mathrm{grid}} \approx 27{,}000$ per pattern at $\tau_s = 0.15\,\mathrm{ms}$) and, as noted, the capacity gap there is inside finite-$N$ noise; we deliberately skipped that corner for the *capacity* test (it is exactly where the *witness* is cheapest and most decisive). The few-percent finite-$T$ Whittle edge correction to $K_{\mathrm{eff}}^{\mathrm{PR}}$ (O2) and the white-afferent assumption (O3) are likewise not quantified here.

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